THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** Quadrant I	Check the signs of the abscissa and ordinate.
2	**(a)** 1	The degree is the highest power of the variable.
3	**(a)** $\sqrt{49}$	A rational number can be written as p/q with $q \neq 0$.
4	**(c)** $x^2 + 4x + 4$	$(a+b)^2 = a^2 + 2ab + b^2$
5	**(b)** 90°	An angle in a semicircle is a right angle.
6	**(d)** 84	Area $= l \times b$.
7	**(b)** $\frac{1}{2}$	$P(E) = \frac{\text{favourable}}{\text{total}}$ with 6 equally likely outcomes.
8	**(c)** 11	Put $n = 5$: $2 \times 5 + 1 = 11$.
9	**(b)** a number is divisible by 10	The part after 'if' is the hypothesis.
10	**(c)** 42	Sum 210 divided by 5.
11	**(c)** 549	Add column by column from the right, carrying 1 when a column sum exceeds 9.
12	**(c)** 100°	Angle sum of a quadrilateral is 360°.
13	**(a)** 2	$-4x + 2y = 6$ becomes $-4x + 2y - 6 = 0$.
14	**(b)** 224 cm^3	$lbh = 4 \times 8 \times 7$.
15	**(d)** 6	The abscissa is the x-coordinate.
16	**(d)** 11	Substitute x and simplify.
17	**(a)** $\sqrt{2}$	The square root of a non-square natural number is irrational.
18	**(d)** $x^2 - 8x + 16$	$(a-b)^2 = a^2 - 2ab + b^2$
19	**(a)** 7	Radius $= \frac{d}{2}$.
20	**(a)** 46	Perimeter $= 2(l + b)$.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Abscissa $= 4$, ordinate $= 3$ (1); Point $= (4, 3)$ in Quadrant I (1).	1+1
22	$p(2) = 3(2) + (5)$ (1); $p(2) = 11$ (1).	1+1
23	Divide 3 by 5 (1); $\frac{3}{5} = 0.6$ (1).	1+1
24	Use $(a+b)^2 = a^2 + 2ab + b^2$ (1); $= x^2 + 16x + 64$ (1).	1+1
25	Angle at the centre is double that at the circumference (1); Required angle $= \frac{36^\circ}{2} = 18^\circ$ (1).	1+1
26	$C = 2\pi r = 2 \times \frac{22}{7} \times 21$ (1); $C = 132$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	$AB = \sqrt{(4-(1))^2 + (6-(2))^2} = \sqrt{(3)^2 + (4)^2}$ (1); $AB = \sqrt{9+16} = \sqrt{25}$ (1); $AB = 5$ units (1).	1+1+1
28	$2x - 10 = 0$ gives $x = 5$ (1); Check: $p(5) = 2(5) + (-10) = 0$ (1); Hence the zero is verified (1).	1+1+1
29	Let $x = 0.\overline{2}$; $10x = 2.\overline{2}$ (1); $10x - x = 2$, so $9x = 2$ (1); $x = \frac{2}{9}$ (1).	1+1+1
30	Use $(a+b)^2 = a^2 + 2ab + b^2$ (1); $a = 2x$, $b = 3$ (1); $= 4x^2 + 12x + 9$ (1).	1+1+1
31	Half-chord $= \sqrt{5^2 - 3^2}$ (1); $= \sqrt{16} = 4$ cm (1); Chord $= 2 \times 4 = 8$ cm (1).	1+1+1
	OR: Area $=$ base $\times$ height (1); $108 = 12 \times h$ (1); $h = 9$ cm (1).	1+1+1
32	Total $= 10$ (1); $P(\text{red}) = \frac{3}{10}$ (1); $P(\text{not red}) = 1 - \frac{3}{10} = \frac{7}{10}$ (1).	1+1+1
	OR: $t_n = a + (n-1)d$ (1); $t_{10} = 3 + 9 \times 4$ (1); $= 39$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	$AB = \sqrt{15^2 + 0^2} = 15$ (1); $BC = \sqrt{0^2 + 20^2} = 20$ (1); $AC = \sqrt{225 + 400} = 25$ (1); $AB^2 + BC^2 = 225 + 400 = 625$ (1); $AC^2 = 625$, so $AB^2 + BC^2 = AC^2$: right angle at B (1).	1+1+1+1+1
34	$F(x) = 10x + 30$ (1); $F(6) = 10 \times 6 + 30$ (1); $F(6) =$ ₹90 (1); $10x + 30 = 110$ (1); $x = 8$ km (1).	1+1+1+1+1
35	$\sqrt{25}$ is rational (1); $\sqrt{6}$ is irrational (1); $\sqrt{36}$ is rational (1); $\frac{3}{7}$ and 0.6 are rational (terminating or fraction) (2).	1+1+1+2
36	Use $(a+b)^2 = a^2 + 2ab + b^2$ (1); $(2x)^2 = 4x^2$, $2(2x)(y) = 4xy$ (1); $= 4x^2 + 4xy + y^2$ (1); At $x = y = 1$: 9 (2).	1+1+1+2
	OR: The perpendicular from the centre bisects the chord (1); Half-chord $= 12$ cm (1); $r = \sqrt{12^2 + 9^2} = \sqrt{225}$ (1); $r = 15$ cm (1); Radius found using the right triangle (1).	1+1+1+1+1
37	Perimeter $= 2(40 + 25) = 130$ m (1); Area $= 40 \times 25 = 1000$ m^2 (1); Cost $= 130 \times 12$ (1); $=$ ₹1560 (1); Units stated (1).	1+1+1+1+1
	OR: Sample space: HH, HT, TH, TT (4 equally likely outcomes) (1); $P(\text{two heads}) = \frac{1}{4}$ (1); $P(\text{at least one head}) = \frac{3}{4}$ (1); $P(\text{no head}) = \frac{1}{4}$ (1); Check: $P(\text{no head}) = 1 - P(\text{at least one head})$ (1).	1+1+1+1+1
38	Terms $4, 7, 10, 13, 16$ (1); Constant difference, so an AP (1); $t_{10} = 4 + 9 \times 3$ (1); $= 31$ (2).	1+1+1+2
	OR: $3 + 5 = 8$, which is even (1); Let the numbers be $2m + 1$ and $2n + 1$ (1); Sum $= 2m + 2n + 2 = 2(m + n + 1)$ (1); A multiple of 2, so even (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** $\frac{1}{2}$	Two equally likely outcomes.
2	**(d)** 3	Subtract any term from the next term.
3	**(d)** its ones digit is 0	The part after 'then' is the conclusion.
4	**(c)** 40	Equal-sized groups: average of the averages.
5	**(b)** 8	Divisors of 24: 1, 2, 3, 4, 6, 8, 12, 24.
6	**(d)** 118°	Opposite angles are equal; adjacent angles add up to 180°.
7	**(c)** Yes	$5(5)+(7)(3)=46$, equal to 46.
8	**(a)** 125 cm^3	5^3.
9	**(c)** -6	The ordinate is the y-coordinate.
10	**(d)** 5	Solve $p(x)=0$.
11	**(b)** 0.75	Divide the numerator by the denominator.
12	**(c)** x^2-9	$(a+b)(a-b)=a^2-b^2$
13	**(d)** 5	The perpendicular from the centre bisects the chord.
14	**(c)** 36	Perimeter $=4\times$ side.
15	**(c)** $\frac{1}{2}$	$P=\frac{\text{green balls}}{\text{all balls}}$.
16	**(d)** 22	Add the common difference 5 to the last term.
17	**(c)** false	Counterexample: 2. 2 is prime and even.
18	**(a)** 87	29% of $300=\frac{29}{100}\times 300=87$.
19	**(a)** 2	$38=12\times 3+2$.
20	**(a)** 44 cm	Opposite sides are equal.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Favourable outcomes $=3$, total outcomes $=6$ (1); $P=\frac{1}{2}$ (1).	1+1
22	$t_1=4$, $t_2=7$ (1); $t_3=10$, $t_4=13$ (1).	1+1
23	Converse: If the ones digit of a number is 0, then the number is divisible by 10. (1); it is true (1).	1+1
24	Sum $=55+27+50+44+54=230$ (1); Mean $=\frac{230}{5}=46$ (1).	1+1
25	Add from the units column, carry 1 when a column sum exceeds 9; Columns with a carry: units (1); $318+427=745$ (1).	1+1
26	Angle sum $=360^\circ$; Sum of the three given angles $=315^\circ$ (1); Fourth angle $=360^\circ-315^\circ=45^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Converse: If the diagonals of a quadrilateral are perpendicular, then it is a rhombus. (1); Counterexample: a kite that is not a rhombus has perpendicular diagonals (1); It satisfies the hypothesis of the converse but not its conclusion, so the converse is false (1).	1+1+1
28	Total of first group $=10\times 47=470$ (1); Total of second group $=30\times 31=930$ (1); Combined average $=\frac{1400}{40}=35$ (1).	1+1+1
29	divisors(10) $=[1,2,5,10]$ (1); divisors(30) $=[1,2,3,5,6,10,15,30]$ (1); Common divisors $=[1,2,5,10]$, greatest is 10 (1).	1+1+1
30	Diagonal $=\sqrt{9^2+12^2}$ (1); $=\sqrt{225}=15$ cm (1); Perimeter $=2(9+12)=42$ cm (1).	1+1+1
31	$x=0$: $6y=36$ gives $y=6$, so $(0,6)$ (1); $y=0$: $4x=36$ gives $x=9$, so $(9,0)$ (2).	1+2

	OR: Volume $= lbh = 5 \times 15 \times 9 = 675$ cm^3 (1); TSA $= 2(lb + bh + hl) = 2(75 + 135 + 45) = 510$ cm^2 (1); LSA $= 2h(l + b) = 360$ cm^2 (1).	1+1+1
32	Signs $(+, +)$: Quadrant I (1); Reflection in x-axis: $(5, -8)$ (1); Reflection in y-axis: $(-5, 8)$ (1).	1+1+1
	OR: $x = 0$: $y = 1$ (1); $x = 1$: $y = 4$ (1); $x = 2$: $y = 7$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Sum of 12 observations $= 12 \times 50 = 600$ (1); New sum $= 600 + 63 = 663$ (2); New mean $= \frac{663}{13} = 51$ (2).	1+2+2
34	Even values: 2, 4, 6, 8, 10, 12, 14 (1); Total $= 2 + 4 + 6 + 8 + 10 + 12 + 14$ (2); Final total $= 56$ (2).	1+2+2
35	Sum of ratio terms $= 12$ (1); One part $= 360^\circ \div 12 = 30^\circ$ (2); Angles: 30°, 90°, 150°, 90° (2).	1+2+2
36	$6y = -7x + 43$ (1); $y = -\frac{7}{6}x + \frac{43}{6}$ (2); Slope $= -\frac{7}{6}$, y-intercept $= \frac{43}{6}$ (2).	1+2+2
	OR: CSA $= 2\pi rh = 2 \times \frac{22}{7} \times 21 \times 5 = 660$ cm^2 (1); TSA $= 2\pi r(r + h) = 3432$ cm^2 (2); Volume $= \pi r^2 h = 6930$ cm^3 (2).	1+2+2
37	Quadrant: Quadrant I (1); Distance from x-axis $= \lvert 4 \rvert = 4$; from y-axis $= \lvert 3 \rvert = 3$ (1); $OP = \sqrt{9 + 16} = 5$ (1); Reflection in the origin: $(-3, -4)$ (1); The origin is the midpoint of P and its image: $\left(\frac{3+(-3)}{2}, \frac{4+(-4)}{2}\right) = (0, 0)$ (1).	1+1+1+1+1
	OR: Slope $= 2$ (1); y-intercept $= 6$ (1); x-intercept: $0 = 2x + (6)$, $x = -3$ (1); At $x = 3$: $y = 12$ (1); So the point lies on the line (1).	1+1+1+1+1
38	Long division gives remainders that repeat (1); $\frac{5}{11} = 0.\overline{45}$ (1); Repeating block 45 (1); Block length $= 2$ (1); The expansion is non-terminating and repeating, so the number is rational (1).	1+1+1+1+1
	OR: Find two numbers with sum 5 and product 6 (1); The numbers are 2 and 3 (1); $= (x + 2)(x + 3)$ (1); Check by expanding (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** 5	$6y = 34 - (2)(2)$ gives $y = 5$.
2	**(b)** 96 cm^2	$6a^2$.
3	**(c)** 5	$\sqrt{3^2 + 4^2} = 5$
4	**(b)** 5	Put $x = 0$: $y = c$.
5	**(d)** $0.\overline{3}$	Divide the numerator by the denominator.
6	**(a)** 961	$31 = (30 + 1)$, apply $(a + b)^2$
7	**(c)** $25°$	The angle at the centre is double the angle at the circumference.
8	**(d)** 48	Area $= \frac{1}{2} \times$ base $\times$ height.
9	**(c)** $\frac{1}{2}$	Count the multiples of the number.
10	**(d)** 16	Multiply the last term by the common ratio 2.
11	**(c)** If a figure is a square, then it is a rectangle.	Subject becomes the hypothesis.
12	**(b)** 95	Total length is the sum of the components.
13	**(b)** 2	Common divisors: 1, 2.
14	**(b)** rhombus	Definition of a rhombus.
15	**(c)** a straight line	Fact.
16	**(d)** 84 cm^2	$2h(l + b)$.
17	**(d)** $(4, 6)$	Average the abscissae and the ordinates.
18	**(a)** 3	In $y = mx + c$ the slope is m.
19	**(d)** terminating	Recall the basic facts.
20	**(b)** 2499	$(50 - 1)(50 + 1) = 2500 - 1$

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	$4x + 8y - 5 = 0$ (1); $a = 4$, $b = 8$, $c = -5$ (1).	1+1
22	Volume $= lbh = 9 \times 6 \times 3 = 162$ cm^3; TSA $= 2(lb + bh + hl) = 2(54 + 18 + 27) = 198$ cm^2 (1); LSA $= 2h(l + b) = 90$ cm^2 (1).	1+1
23	$d = \sqrt{6^2 + 8^2} = \sqrt{100}$ (1); $d = 10$ units (1).	1+1
24	Put $3x - 15 = 0$ (1); $x = 5$ (1).	1+1
25	$\sqrt{16} = 4 = \frac{4}{1}$ (1); So it is rational (1).	1+1
26	Use $(a + b)(a - b) = a^2 - b^2$ (1); $= x^2 - 64$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Divide 5 by 12 (1); $\frac{5}{12} = 0.41\overline{6}$ (1); Repeats: block 6 (1).	1+1+1
28	Compare with $a^2 + 2ab + b^2$ (1); $a = x$, $b = 3$ (1); $= (x + 3)^2$ (1).	1+1+1
29	Angle at the centre is double that at the circumference (1); $\angle ACB = 24°$ (1); Angles in the same segment are equal (1).	1+1+1
30	$s = \frac{3+4+5}{2} = 6$ (1); $A = \sqrt{s(s-a)(s-b)(s-c)}$ (1); $A = 6$ cm^2 (1).	1+1+1
31	$P(4) = \frac{22}{120} = \frac{11}{60}$ (1); $P(\text{not } 4) = 1 - \frac{11}{60}$ (1); $= \frac{49}{60}$ (1).	1+1+1
	OR: $S_n = \frac{n(n+1)}{2}$ (1); $S = 20 \times 21 \div 2$ (1); $= 210$ (1).	1+1+1
32	A false statement needs only one counterexample (1); Take $x = -4$ (1); $16 > 9$ but $-4 < 3$, so the statement is false (1).	1+1+1

	OR: Sum of weekly averages $= 123$ (1); Average $= \frac{123}{3} = 41$ min per day (1); Every week has the same number of days, so each average has the same weight (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Angle in a semicircle is 90° so $\angle ACB = \angle ADB = 90^\circ$ (1); $\angle ABC = 90^\circ - 25^\circ = 65^\circ$ (1); $\angle ABD = 90^\circ - 30^\circ = 60^\circ$ (1); $\angle CBD = 125^\circ$ is not needed here (1); Sum check: $\angle ABC + \angle ABD = 125^\circ$ (1).	1+1+1+1+1
34	$C = 2\pi r = 88$ m (1); $A = \pi r^2 = 616$ m^2 (1); Cost $= 88 \times 15$ (1); $=$ ₹1320 (1); Units stated (1).	1+1+1+1+1
35	Total outcomes $= 6$ (1); $P(\text{an even number}) = \frac{3}{6} = \frac{1}{2}$ (1); $P(\text{a number greater than } 4) = \frac{2}{6} = \frac{1}{3}$ (1); Favourable for 'or': 4 outcomes (1); $P = \frac{4}{6} = \frac{2}{3}$ (1).	1+1+1+1+1
36	$S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S = 10/2 \times [4 + 27]$ (1); $= 5 \times 31$ (1); $S_{10} = 155$ (2).	1+1+1+2
	OR: $6 \times 7 = 42$, even (1); Of two consecutive integers one is even (1); So the product has the factor 2 (1); Hence even (2).	1+1+1+2
37	Pure part in the first $= 20 \times \frac{40}{100} = 8$ mL (1); Pure part in the second $= 16.8$ mL (1); Total $= 24.8$ mL in 80 mL (1); Concentration $= 31\%$ (2).	1+1+1+2
	OR: divisors(50) $= [1, 2, 5, 10, 25, 50]$ (1); divisors(75) $= [1, 3, 5, 15, 25, 75]$ (2); Common divisors $= [1, 5, 25]$, greatest is 25 (2).	1+2+2
38	Diagonals bisect each other at right angles (1); Side $= \sqrt{6^2 + 8^2} = 10$ cm (1); Perimeter $= 4 \times 10 = 40$ cm (1); Area $= \frac{1}{2} \times 12 \times 16 = 96$ cm^2 (2).	1+1+1+2
	OR: $y = 11x + 31$ (1); For $x = 4$: $y = 11 \times 4 + 31$ (2); Fare ₹75 (2).	1+2+2

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